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Editor-in-Chief
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Dear Editor,

I am pleased to submit for your consideration the manuscript

**“Real quadratic discriminants with integer Hurwitz ratio $D^2/B_{2,\chi_D}$:
a finite list of seven”**

for publication in the *Journal of Number Theory*.

The paper is a short note (8 pages) addressing a question that, to the author’s best literature review, has not previously been stated and settled in this exact form. Specifically, for a positive fundamental discriminant D and the associated even Kronecker character χ_D , the rational number

$$k(D) := \frac{D^2}{B_{2,\chi_D}} = \frac{\pi^2 \sqrt{D}}{L(2, \chi_D)} = \frac{D^2}{24 \zeta_K(-1)}, \quad K = \mathbb{Q}(\sqrt{D}),$$

is shown to be an *integer* for *exactly seven* values $D \in \{8, 12, 24, 28, 60, 76, 156\}$, with $k(D) \in \{32, 36, 48, 49, 75, 76, 117\}$ respectively, when ranging over all fundamental $D \leq 10^6$.

The closed form for $L(2, \chi_D)$ is the classical Hurwitz formula (Hurwitz 1882, Iwasawa 1972, Washington 1997), and the values $24\zeta_K(-1)$ are accessible via Zagier’s 1976 tables. What appears to be new is:

- *Explicit normalisation and classification by integrality of $k(D) = D^2/B_{2,\chi_D}$* , rather than by integrality of $24\zeta_K(-1)$ alone. This refinement isolates a strictly smaller set than the one “ $24\zeta_K(-1) \in \mathbb{Z}$ ” that is classically known to be infinite (true for all fundamental D).
- *The explicit finite list of seven elements and its computational verification up to $D = 10^6$* , together with a heuristic density argument ($\sim D^{-3/2}$) supporting completeness.
- *The Hirzebruch–Zagier signature interpretation*, identifying the seven discriminants as exactly those for which the arithmetic part of the Hilbert modular surface signature divides D^2 .
- *The parity-dual observation* that the imaginary-quadratic analogue at $s = 3$ produces exactly one integer- k case ($d = -4$), the simplest odd-character mirror of the seven-element real list.

The proof of the seven-element classification is short: a direct evaluation of B_{2,χ_D} for each of the seven values, together with an exhaustive sweep verified by two independent methods (direct Bernoulli sum and `lfun`-based functional-equation evaluation). The verification protocol is reproduced inside the paper for the reader’s convenience.

Manuscript length: 8 pages, including 1 table, 1 numerical appendix and 19 references.

No conflict of interest. The author has no financial or non-financial conflicts to disclose.

Suggested reviewers: scholars who have published on $\zeta_K(-1)$ for real quadratic fields, or on Hilbert modular surface signature, will be particularly well placed to evaluate this work. Suggestions (declared without solicitation):

- Don Zagier (Max-Planck-Institut für Mathematik, Bonn),
- Lawrence C. Washington (University of Maryland),
- Gerard van der Geer (Universiteit van Amsterdam).

Status: original submission, not under review elsewhere.

I am grateful for your consideration.

Sincerely,

Kévin Remondière